Blockchain based Electronic Voting System for Secure and Transparent Elections in India

Sahil Hussain[a], Harsh Deep[b], Aadhyaa Pahariya[c], Prabhat Kumar Maurya[d], Pushpa Singh[e]

[a,b,c,d,e]Computer Science & Engineering, G.L. Bajaj Institute of Technology and Mangement, Greater Noida, 201306, India
[a]sahilhusain790@gmail.com, [b]harshdeepgupta2002@gmail.com, [c]aadhyaapahariya7705@gmail.com, [d]prabhatkm2022@gmail.com, [e]pushpa.gla@gmail.com

Abstract:
Blockchain's technologies have the potential to revolutionize the voting system by introducing greater transparency, security, immutability and trust in the electoral process. Traditional voting methods often face challenges such as vote tampering, low voter turnout, and lack of verifiability. This paper discusses the security of the electoral process, addressing problems like voter impersonation, hacking, and tampering of votes. The aim of this paper is to discuss the implementation of blockchain technologies in the existing voting system. The proposed blockchain-based electronic voting system (BEVM) has two modules: Admin and User. Admin modules can create a new election, add the candidates, and count the votes. The user module can see the list of candidates contesting in elections and cast their vote. The voting process is carried out as a smart contract and processed over the Ganache Blockchain Server. The user casts their vote using their MetaMask account, and the Admin system verifies that the transaction has occurred. The blockchain, with its decentralized ledger, allows for transparency and security of election results. The immutability characteristic assures the integrity of the voting process by eliminating the chance of tampering.

Keywords—Blockchain, transparency, decentralization and E-voting.

1. Introduction

India is the world's largest democratic Republic country and elections are essential and fundamental to the democratic process. Since 1982, India's elections have been mainly conducted using the Electronic Voting Machine (EVMs) which makes the voting and the counting process efficient and accurate. Though paper ballots are also in use in some cases. One of the key problems with EVM is the concern about potential tampering, hacking, technical malfunction, etc. In 2019, VVPATs were used alongside EVMs in all polling booths across the country for additional security. However, several cases are reported for the EVM hacking, manipulation, and capturing (Banerjee et al., 2021).

The security and integrity of elections become essential for a technologically dependent nation like India with millions of people. As the country transforms into a thoroughly digitized one, there is an immediate necessity to embrace frontier technologies that can ensure elections' integrity and regain the people's trust in the democratic system. Blockchain technology is a decentralized digital ledger that provides immutability, transparency, security, trust, and integrity (Singh et al., 2023).

The use of Blockchain is very promising if implemented for voting systems. Blockchain decentralizes the act of voting is recorded into an immutable distributed ledger built in for every vote. Once you have voted, your vote may not be retracted or removed. This decentralization tends to curtail the risks that are associated with control in a central system that is a gigantic problem in current voting infrastructure used in India.

In addition, blockchain-based voting systems offer public auditability which makes it possible for third-party verification of votes to increase transparency without revealing the identity of the voters. This transparency fosters public trust, voters can verify that their votes have indeed been correctly tallied. It's a form of transparency that could also serve as an anti-fraud feature since unauthorized attempts at altering previous voting records will be relatively impossible (Hossain et al., 2024). This innovation may make tampering or manipulation of votes less likely, and thus, there is an enhancement of the credibility of elections in India as a whole.

The existing electoral process is particularly inaccessible to locations like rural areas or people with disability, and forms an important feature of blockchain-based voting systems. These methods of voting have been cumbersome with regard to providing long lines and polling stations that are difficult to access, which discourages the participation of potential voters. With safe online voting, voter turnout will be increased, and the process for holding elections will become more democratic. It becomes even more crucial in a vast country like India, where the participation of voters must be very high to ensure that democracy remains stable. Blockchain makes the voting system more accessible to ensure all the eligible persons and offers a secure, authenticated voting system without the need for tallying authorities to verify and finalize election results (Jayakumari et al., 2024).

The second benefit is the scalability of the blockchain voting platforms, whereby it can be applied in large-scale national and local Indian elections (Hajian et al, 2023). Due to it being both scalable and secure, coupled with the fact that it offers increased transparency than any other system, it would ensure fair elections in India. Thus, Blockchain technology is the cure for all the underlying issues of the current traditional voting systems, which need to be explored for the implementation of the current electoral process. This paper uncovers the suitability of Blockchain, which can address a few vulnerabilities of the existing voting system faces when implemented into an open, secure, and transparent election mechanism so as to make electoral integrity possible.

The objective of this paper is to design a scalable blockchain-based electronic voting system (BEVS) that ensures security, transparency, and privacy, using smart contracts for accurate vote counting, decentralized control techniques for safeguarding voter identities. Major contributions

Electronic copy available at: https://ssrn.com/abstract=5770662

of this paper are:

- Review and analysis of existing e-voting system for election.
- Secure and transparent system of votes on a blockchain via smart contracts.

The remainder of the paper is structured as follows: Section 2 literature survey work, including e-voting and previous e-voting services worldwide, e-Voting system requirements, and the blockchain based e-voting system. Section 3 presents the proposed work including the admin zone, user zone, database and various diagrams of the Blockchain based e-voting System. Section 4 presented the method to incorporate security and transparency in the proposed system. Section 5, explore the result and discussion followed by anticipated impact of system i.e. discussed in the section 6. In the last section concludes the paper along with future work.

2. Literature Survey

The e-voting systems of today are around the globe and, indeed in India, have different features with creative innovation but come with multiple issues (Banerjee et al., 2021; Jafar et al., 2021). Blockchain technology ensures anonymity and transparency as a voter can confirm whether their vote was captured accurately without disclosing their identity (Alvi et al., 2022) [6] and provides scalability (Hajian et al, 2023). Johnson in 2019 pointed out the potential of blockchain-based voting in a manner that combats the risks associated with the technology and strengthens public confidence in electoral and political processes and secures a balanced system of governance in the US and the EU constitutional orders. Also, homomorphic encryption will be applied to vote privacy and thus allow computation on encrypted data; therefore, this ensures that votes can be tallied without their decryption so as to maintain the anonymity of the voter even through the process (Zhan et al., 2024). It is also indicated in countries like Africa, where blockchain decentralized architecture provides flexibility and security to the entire process. Flexibility in terms of voter access at convenient locations and fully secure on the blockchain network layer (Alam, Yusuf, & Sani, 2020). Shevtekar, Kumar, & Sathe in 2024 presented a secure and transparent system for the trustworthy electoral process by using blockchain technology but they have not highlighted how the process is secure. Further, Kafhali (2024) discussed an end-to-end electronic voting system based on blockchain, with the help of cryptographic techniques such as zero-knowledge proofs to improve privacy. The presented study reflected the contributions of blockchain in the voting system, but still, more exploration is required for real implementation.

Estonia's i-Voting comes with digital ID verification and real-time audits but is pricey to be installed and troubled by privacy issues (Vassil & Solvak, 2016). Voatz uses a mobile blockchain system with biometric authentication, which was the first internet voting application used in US- federal elections (Specter, Koppel, & Weitzner, 2020), still having security flaws and limited accessibility. Similarly, SecureVote also relies on decentralized and blockchain technologies, but still has challenges with user acceptability, expensive transactions, and risks from changes in protocols of blockchain (Xu et al., 2029).

India's blockchain voting projects, such as Telangana and ECI, show potential, but they confront cost, privacy, and scalability issues (Telangana Government, 2020). IIT Madras also proposed a technology named as blockchain based Remote voting in India but it is still in pilot phase (The Hindu, 2021). The table I, highlights some existing e-voting systems, including India's initiatives, and their research gaps.

Table 1 – Existing e-voting systems

References	Voting System	Research Gap
Vassil, et al., (2016)	Estonian i-Voting	High Infrastructure Costs, privacy concerns
Specter, Koppel, & Weitzner,, et al., (2020)	Voltz	Security vulnerabilities, limited tech accessibility
Weber, et al., (2019)	SecureVote	Not discussed the particular aspects of EVM of any country
Telangana Govt. (2020)	Blockchain Voting (Telangana, India)	High Infrastructure cost limited to Local Point
ECI & IIT Madras (2021)	Blockchain-based Remote Voting (India)	Still in Pilot phase, privacy and scalability concerns

Therefore, it can be said that there have been many blockchain-based voting systems globally and also in India which showcases a bit of potential but still pretty much the same challenges over the years solving those drawbacks where traditional methods are restricted. These included high development costs, privacy intrusions, security bugs and scalability problems.

To address these and other associated challenges, this study presents a secure, scalable, and cost-effective blockchain-based electronic voting system (BEVS). This paper aims to build a system using ganache and metamask to provide a more efficient, user-friendly, and scalable solution that eliminates such persistent problems and encourages secure elections across the nation. The system is implemented in react.js.

3. Proposed Work

The proposed blockchain-based voting system is designed to address the challenges of traditional voting methods by leveraging the transparency, security, and immutability features offered by blockchain technology. The system has two main user roles: Admin and User. Each role has specific functionalities, which are detailed below.

3.1. Admin Interface

The home interface presents two options: Admin and User. When the Admin option is selected, the user is redirected to a login page. For verification purposes, the admin must enter a username and password. Once authenticated, the admin gains access to the Admin Zone. Within this zone, the admin is provided with the following functionalities:

- Election Creation: The admin can initiate the creation of a new election, specifying the election details such as title, description, and available candidates.
- Add Candidates: Through integration with MetaMask, the Admin can add candidates to the election. This integration ensures secure

Electronic copy available at: https://ssrn.com/abstract=5770662

transactions, as MetaMask allows interaction with the blockchain via its wallet and Ethereum network.

- View Vote Counts: The admin has the ability to check the ongoing vote counts for the elections. The votes are recorded as blockchain transactions, providing an immutable record of the voting process.

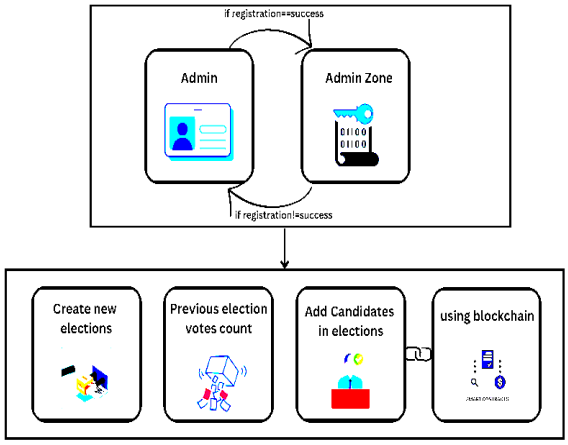


Fig.1. Pictorial representation of the admin zone working.

3.2 User Interface

When the User option is selected from the home interface, the user is directly taken to the Elections Module, where they can view all available elections. The user can browse the elections, view detailed information about each election, and see the list of candidates contesting in those elections. The frontend of the project is done with react.js.

Voting Process: Once the user selects an election, they can choose a candidate and cast their vote. The voting process is carried out through a smart contract, which ensures that each vote is securely recorded on the blockchain.

Blockchain Integration: The vote is processed as a transaction on a private Ganache Blockchain Server. Each vote, represented as a single transaction, is recorded, ensuring transparency and immutability. After the user casts their vote using their MetaMask account, the system verifies that the transaction has occurred.

Vote Limitation: To prevent double voting, the system ensures that once a vote is cast, it cannot be changed or repeated. This limitation is enforced by the blockchain's immutable nature, which guarantees that a user cannot vote multiple times for the same or a different candidate. The blockchain serves as the merit of the system by enforcing this one-vote rule. Fig.2 illustrates the graphical representation of the user interface for the proposed user zone.

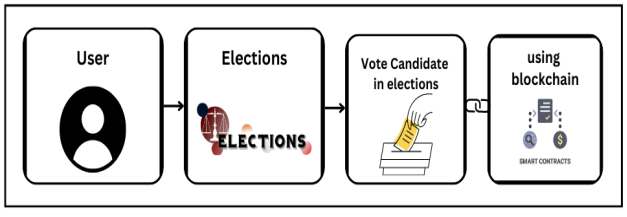


Fig. 2 - Pictorial representation of the user zone working.

3.3 Data Storage:

The system uses MongoDB, a NoSQL database, to store non-blockchain data related to the elections. The ethereum blockchain platform is used to develop smart contracts that provide secure vote handling. MongoDB is essential for maintaining non-transactional data such as election details and administrative credentials. The separation of tasks between blockchain and MongoDB improves the system's efficiency and scalability. Furthermore, MetaMask maintains voter legitimacy and fairness by requiring wallet-based authentication, which prohibits multiple votes from the same wallet. The following data is stored in MongoDB.

Election Details: Information about each election, including the election name, description, date, and the list of candidates, is stored securely in MongoDB.

Admin Credentials: The system stores admin credentials in MongoDB for user verification, ensuring that only authorized individuals can create elections, add candidates, and access election results. By using MongoDB, the system can quickly retrieve election-related data without accessing the blockchain, which is designed to store transactional data.

Use Case diagram of proposed work for Blockchain- E-Voting System (BEVS) is shown in Fig. 3.

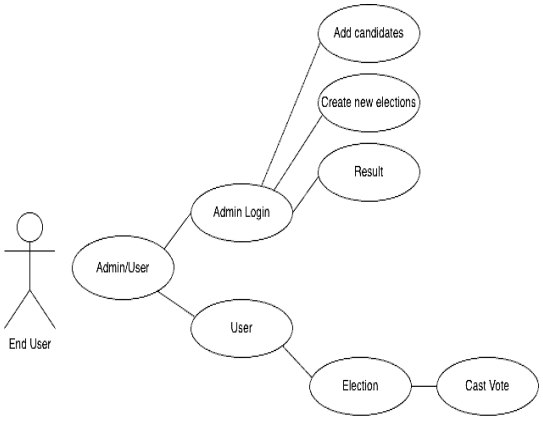


Fig. 3- Use Case diagram of Proposed work for BEVS

Blockchain and Ganache Event Handling:

Each vote is treated as a unique transaction in the Ganache Blockchain. After a user casts a vote, a transaction is initiated and recorded on the blockchain. For every vote transaction, Ganache triggers an event that notifies the system of the completion of the vote. This event mechanism serves as a way to:

Track Vote Transactions*:* Every time a vote is cast, an event is raised in the Ganache server, allowing the system to track and confirm the transaction. Ensure Transparency: These events help provide real-time updates on vote counts and ensure that each vote is accounted for properly. The transparency of the blockchain ensures that the election results can be audited at any time.

The blockchain ensures the system's integrity by storing vote data in an immutable ledger. Each vote transaction, once confirmed, is recorded permanently and cannot be altered. The integration of smart contracts automates the voting process, ensuring that votes are cast securely and accurately. Moreover, by using MetaMask for user authentication, the system ensures that only verified users can participate in the voting process.

Electronic copy available at: https://ssrn.com/abstract=5770662

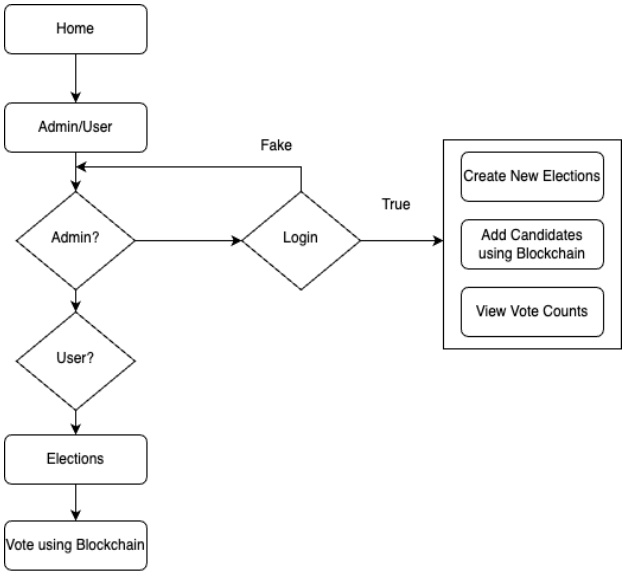


Fig. 4 - Proposed work for Blockchain Based E-Voting System

4. Transparency and Security in BEVS

Transparency is critical in democratic voting systems, yet traditional electronic and paper-based procedures frequently lack visibility, causing trust concerns. Blockchain tackles this issue by creating an open, immutable ledger that records every action—voter authentication, candidate listing, vote casting, and counting—as timestamped, digitally signed transactions over a decentralized network. This allows stakeholders and auditors to verify the election's integrity while protecting voter privacy. Smart contracts increase transparency by automatically enforcing rules, such as prohibiting multiple voting and guaranteeing that only eligible users can vote, removing the need for manual oversight.

Security is critical in electronic voting systems to prevent assaults that threaten democratic integrity. Blockchain provides a solid security basis via cryptography, decentralization, and consensus procedures. Immutability assures that once recorded, a vote cannot be changed, preventing tampering. Public-key cryptography secures votes, protects voter anonymity, and ensures data integrity. Unlike centralized systems, blockchain's decentralized nature distributes data across nodes, lowering the likelihood of attacks and data loss. Consensus algorithms validate transactions, allowing only legitimate votes to enter the ledger. Algorithm 1 presented the steps to provide voter registration with authentication, while secure vote casting is represented in algorithm 2, and algorithm 3 is for counting the vote.

Algorithm-1: Voter Registration and Authentication

```
Input: Voter_ID, MetaMask_Address
Output: Confirmation message
BEGIN
  IF Voter_ID is valid THEN
    Hashed_ID ← Hash(Voter_ID)
    Store (Hashed_ID, MetaMask_Address) on blockchain using smart
    contract
    RETURN "Voter successfully registered"
  ELSE
    RETURN "Registration failed: Invalid Voter ID"
END
```

Algorithm-2: Voter Casting

```
Input: Encrypted_Vote, Voter_Private_Key
Output: Confirmation of vote recording or rejection
BEGIN
  Voter selects candidate and generates Encrypted_Vote
    Signed_Vote ← Sign(Encrypted_Vote, Voter_Private_Key)
    Submit Signed_Vote to blockchain network
  Blockchain nodes perform verification:
  IF Signature is valid AND Voter has not voted before THEN
      Append Signed_Vote to blockchain
      RETURN "Vote successfully recorded"
    ELSE
      RETURN "Vote rejected: Invalid signature or duplicate vote"
  END
```

Algorithm-3: Vote Tallying (Pos-Election):

```
  Input: Blockchain (Encrypted_Votes)
Output: Final vote count per candidate
BEGIN
    Encrypted_Votes ← Fetch all vote transactions from blockchain
    Vote_Counts ← Aggregate votes by candidate
    Publish Vote_Counts to blockchain
    RETURN "Final vote tally successfully published"
  END
```

5. Result and Discussion

The result of this study has shown benefits in addressing important issues connected with traditional voting techniques, such as security, transparency, and voter integrity. The system successfully integrates MongoDB for dynamic data management with Ganache, a blockchain

Electronic copy available at: https://ssrn.com/abstract=5770662

platform for vote recording, to ensure that all transactions are safely kept and verified as shown in the fig.5.

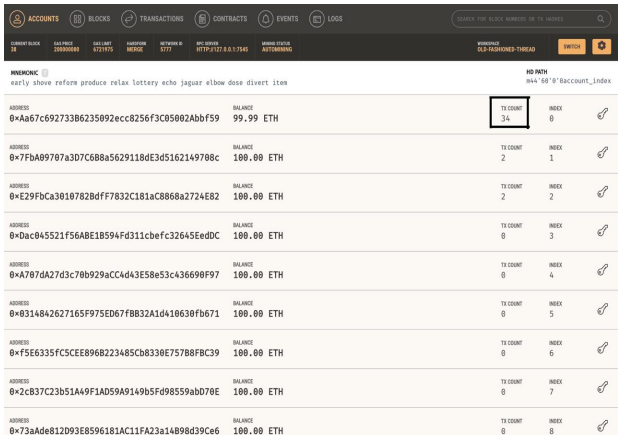


Fig. 5 - Transaction counts by using Ganache

The results show that using smart contracts to automate the voting process works perfectly. Voters could use MetaMask to cast their votes, and each transaction is saved as an immutable entry in the blockchain. This feature prevents post-submission vote changes and assures that each user is only permitted one vote, as enforced through wallet-based verification. Test simulations on the Ganache server revealed that each vote resulted in an event, which confirmed the transaction and allowed for real-time monitoring of the voting process. The number of elections is listed in the admin zone, and the admin can add candidates in a particular election and can view the vote count as mentioned in Fig. 6.

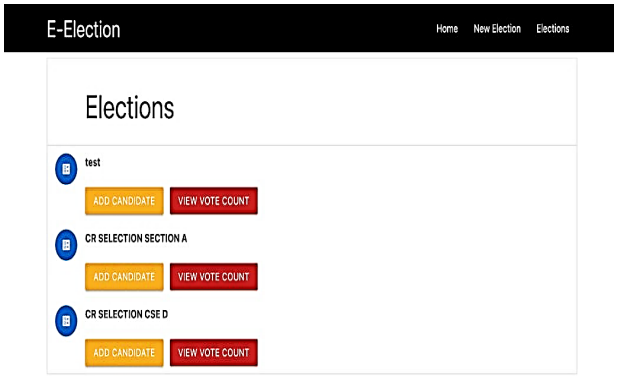


Fig. 6 - Listed existing elections

The results of the voters are seen in the admin zone with counts of the votes with the candidate's name as shown in the fig.7.

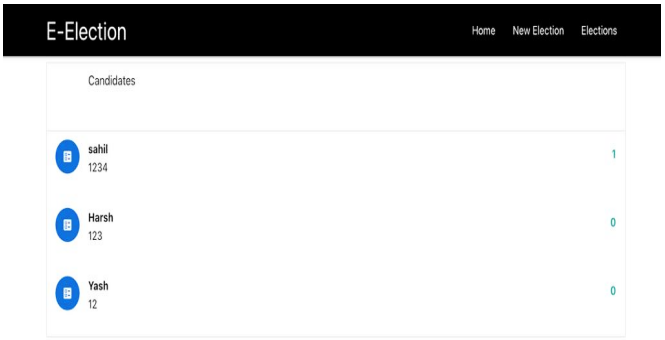


Fig. 7 - Vote counts according to the candidate's name

During testing, the system demonstrated excellent reliability, with no instances of duplicate voting or unauthorized access. However, issues such as user familiarity with MetaMask and limits in the blockchain infrastructure were identified, identifying potential for future improvement. Furthermore, Ganache's event-based architecture enabled real-time feedback, increasing the system's responsiveness and transparency.

Further, Table 2 represents the comparative analysis of traditional, electronic, and blockchain-based voting systems.

Table 2 – Comparative analysis of voting systems

Feature	Traditional Voting	Electronic based voting	Blockchain Based voting
Security	Prone to tampering, ballot fraud	Vulnerable to software/hardware hacks	Cryptographically secure; immutable records
Transparency	Limited; requires observers	Limited; internal process hidden	Full transparency via public ledger
Vote Privacy	High (paper secrecy)	Moderate	High via encryption and anonymity
Auditability	Manual recount possible	Difficult and dependent on system logs	End-to-end verifiable, open audit trails
Speed of Results	Slow (manual counting)	Faster than manual counting	Near real-time tallying with transparency
Cost	High due to paper, logistics	Moderate (hardware dependent)	Initially high, reduces with reuse and scale

6. Impacts, Benefits and Challenges

The implementation of blockchain technology in India's e-voting system could bring significant impacts, such as enhancing the transparency,

Electronic copy available at: https://ssrn.com/abstract=5770662

security, and efficiency of the electoral process. Below are some key points shown in Fig 8.

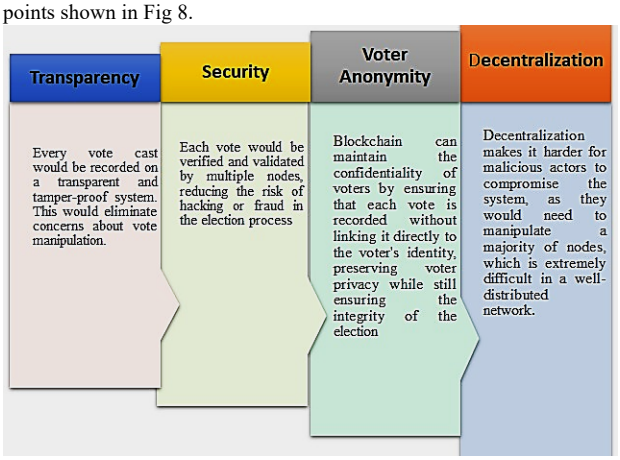


Fig. 8- Impact of BEVS on the existing voting system

Fig. 8 represented the benefits of the proposed system for society. The potential benefits of blockchain implementation could significantly strengthen the integrity of India's democratic process. Proposed methodology offers many benefits such as Enhanced Trust and Credibility, cost efficiency, remote voting, and reducing fraudulent votes, duplicate votes.

However, the system may face scalability challenges when implementing a blockchain-based electronic voting system for large populations

7. Conclusion

The blockchain-based voting system proposed in this paper represents a significant step forward in modernizing election administration by tackling fundamental issues such as voter fraud, multiple voting, and a lack of transparency. By integrating blockchain's immutability with MongoDB's efficient data storage and MetaMask's safe authentication, this system offers a comprehensive and dependable solution for conducting secure and transparent elections. The blockchain protects the integrity of the voting process by recording each vote as an immutable transaction, preventing tampering or manipulation. Smart contracts automate the voting process, assuring accurate results and preventing unwanted access. Ganache's integration as a private Ethereum blockchain server allows for real-world testing and deployment, which improves system stability. Each transaction in Ganache generates an event, allowing for real-time vote tracking and reducing the delays often associated with manual vote counting.

While the system has tremendous benefits, issues such as blockchain infrastructure needs and user familiarity with MetaMask highlight topics for further investigation. Nonetheless, this method offers a significant step toward more secure, transparent, and efficient elections, creating a solid platform for widespread deployment in a variety of electoral settings.

8. References

Banerjee, S., Philipose, P., Dayal, J., Burra, S., & Devasahayam, M. G. (2021). Citizens' Commission on Elections' Report on EVMs and VVPAT.

Singh, P., Dash, S., Gupta, A. K., Singh, N., & Yadav, P. D. (2023, December). Utilizing blockchain for effective urban street vendor management. In 2023 4th International Conference on Computation, Automation and Knowledge Management (ICCAKM) (pp. 1–5). IEEE.

Hossain Faruk, M. J., Alam, F., Islam, M., & Rahman, A. (2024). Transforming online voting: a novel system utilizing blockchain and biometric verification for enhanced security, privacy, and transparency. *Cluster Computing*, *27*(4), 4015-4034.

Jayakumari, B., Sheeba, S. L., Eapen, M., Anbarasi, J., Ravi, V., Suganya, A., & Jawahar, M. (2024). E-voting system using cloud-based hybrid blockchain technology. Journal of Safety Science and Resilience, 5(1), 102–109.

Hajian Berenjestanaki, M., Barzegar, H. R., El Ioini, N., & Pahl, C. (2023). Blockchain-based e-voting systems: a technology review. Electronics, 13(1), 17.

Jafar, U., Aziz, M. J. A., & Shukur, Z. Blockchain for electronic voting system—review and open research challenges. Sensors, 21(17), 2021 5874.

Alvi, S. T., Uddin, M. N., Islam, L., & Ahamed, S. (2022). DVTChain: A blockchain-based decentralized mechanism to ensure the security of digital voting system. Journal of King Saud University - Computer and Information Sciences, 34(9), 6855–6871.

Johnson, D. (2019). Blockchain-based voting in the US and EU constitutional orders: a digital technology to secure democratic values? European Journal of Risk Regulation, 10(2), 330–358.

Zhan, Y., Zhao, W., Zhu, C., Zhao, Z., Yang, N., & Wang, B. (2024). Efficient electronic voting system based on homomorphic encryption. Electronics, 13(2), 286.

Alam, M., Yusuf, M. O., & Sani, N. A. (2020). Blockchain technology for electoral process in Africa: a short review. International Journal of Information Technology, 12(3), 861–867.

Shevtekar, S., Kumar, A., & Sathe, M. (n.d.). Blockchain-based electronic voting system: Secure and transparent implementation for trustworthy electoral processes.

El Kafhali, S. (2024). Blockchain-Based Electronic Voting System: Significance and Requirements. Mathematical Problems in Engineering, 2024(1), 5591147.

Vassil, K., & Solvak, M. (2016). E-voting in Estonia: Technological diffusion and other developments over ten years (2005–2015). Johan Skytte Institute of Political Studies, University of Tartu. Retrieved from https://skytte.ut.ee/sites/default/files/2023-2/E_voting_in_Estonia_Solvak_Vassil_2016.pdf

Specter, M. A., Koppel, J., & Weitzner, D. (2020). The ballot is busted before the blockchain: A security analysis of Voatz, the first internet voting application used in US federal elections. In 29th USENIX Security Symposium (USENIX Security 20) (pp. 1535–1553).

Xu, X., Weber, I., & Staples, M. (2019). Case study: SecureVote: Taking a DApp from MVP to production. In Architecture for Blockchain Applications (pp. 257–278).

Telangana Government. (2020). Blockchain-based voting system: Pilot project for local elections. Retrieved from https://it.telangana.gov.in/wp-content/uploads/2022/12/Telangana-Blockchain-Framework.pdf

The Hindu. (2021). Election Commission working with IIT-Madras on e-voting. Retrieved from https://www.thehindu.com/news/national/election-commission-working-with-iit-madras-on-e-voting/article34170161.ece

Electronic copy available at: https://ssrn.com/abstract=5770662